# LOGANATHAN S

Full Stack AI Engineer | MERN Stack
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## PROFESSIONAL SUMMARY

Full Stack AI Engineer with 3+ years of experience in full-stack web development using React.js, Node.js, Express.js, and MongoDB, including hands-on AI integration experience. Skilled in RESTful APIs, JWT authentication, role-based access control, and responsive interfaces. Integrated Google Gemini API, model fallback handling, and Firebase Realtime Database caching to deliver contextual programming guidance.

## TECHNICAL SKILLS

- **Frontend:** React.js, Next.js, Redux Toolkit, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Material UI, Responsive Design
- **Backend:** Node.js, Express.js, RESTful APIs, Middleware, MVC Architecture, WebSockets, Zod
- **Databases & ORM:** MongoDB, MongoDB Atlas, Firebase, PostgreSQL, Redis, Prisma, Mongoose, Schema Design, Indexing
- **AI Integration:** Google Gemini API, OpenAI API, Claude API, Prompt Engineering, LLM Integration, AI Agent Development, Model Fallback Handling
- **Security:** JWT Authentication, Role-Based Access Control (RBAC), Request Validation
- **Testing:** Jest, React Testing Library, Cypress
- **Tools:** Git, GitHub, Postman, VS Code, Cursor AI, GitHub Copilot, Debugging, Performance Optimization
- **Third-Party APIs:** Google Maps API, OpenStreetMap, Razorpay
- **Deployment:** Docker, Linux, Bash, Netlify, Render
- **Currently Learning:** LLM Tool Calling, Retrieval-Augmented Generation (RAG), LangChain.js

## PROFESSIONAL EXPERIENCE

**Full Stack AI Engineer** Feb 2026 – Present
*Coderead Software Training Academy Private Limited*

- Developed and enhanced CookDCode, an AI-powered coding education platform that guides students through programming problem-solving using a structured, four-tier evaluation pipeline.
- Integrated Google Gemini API with model fallback handling to generate context-aware programming hints and support interactive, Socratic-style student guidance.
- Implemented Firebase Realtime Database caching to store and retrieve previously generated hints, reducing redundant AI API calls for similar queries.
- Built rule-based validation logic for 13 conditional programming problems, incorporating boundary-value checks and mathematical misconception detection.
- Combined input handling, rule-based validation, cached hint retrieval, and dynamic AI generation to deliver responsive and personalized learning assistance.

**Full Stack Developer** May 2023 – Jan 2026
*DNASkew Analytics Pvt Ltd*

- Developed and maintained MERN web applications using React.js, Node.js, Express.js, and MongoDB.
- Built responsive UIs with reusable React components, improving code maintainability and consistency.
- Designed RESTful APIs with JWT authentication, role-based access control, request validation, and centralized error handling.
- Implemented interactive dashboards and data modules with React.js and backend API integration.
- Optimized MongoDB queries and indexes to improve API response times and data retrieval efficiency.
- Improved frontend responsiveness through lazy loading, memoization, and code splitting.
- Collaborated with team members on debugging, feature development, and application testing.

## PROJECTS

**CookDCode – AI-Powered Student Evaluation & Coding Guidance Platform**
*Full Stack AI Engineer* *Feb 2026 – Present*

- Designed a four-tier evaluation pipeline comprising greeting handling, rule-based validation, cached hint retrieval, and dynamic AI-generated guidance.
- Developed customized validators for 13 conditional programming problems, including voting eligibility, odd/even, positive/negative numbers, and boundary-condition checks.
- Integrated Gemini 2.5 Flash with fallback to Gemini 3.1 Flash Lite to handle novel student queries and generate contextual programming hints.
- Implemented a Socratic tutoring approach that encourages logical reasoning through progressive hints and guiding questions rather than directly revealing complete code solutions.
- Configured Firebase Realtime Database to persist and reuse previously generated hints for similar queries, improving response efficiency and minimizing repeated AI requests.

**Tech Stack:** Google Gemini API, Firebase Realtime Database, Rule-Based Validation, Prompt Engineering

**VNetPops – Network Visualization Platform** GitHub
*MERN Stack* *May 2023 – Jan 2026*

- Created reusable React.js components for interactive dashboards and data visualization.
- Built RESTful APIs using Node.js and Express.js to retrieve and manage application data.
- Implemented JWT authentication and role-based access control to protect application features.
- Developed modules including Billing and Summary using React.js, RESTful APIs, and MongoDB.
- Integrated frontend components with backend APIs and MongoDB collections.
- Visualized network devices and their relationships through interactive React components.

**Tech Stack:** React.js, Node.js, Express.js, MongoDB, JWT, Tailwind CSS, Cypress

**MatchHub – Live Sports Score Management System** GitHub | Live
*MERN Stack*

- Built a live sports score platform for managing tournaments, teams, players, and match schedules.
- Developed RESTful APIs to manage and retrieve live sports scores using Node.js and Express.js.
- Implemented JWT authentication and role-based dashboards for 4 user roles: administrators, organizers, referees, and teams.
- Optimized MongoDB models and REST APIs to improve score retrieval and tournament performance.

**Tech Stack:** React.js, Node.js, Express.js, MongoDB, JWT, Tailwind CSS

**Stepora – AI Powered Shoe E-Commerce** GitHub | Live
*MERN Stack*

- Built a MERN stack shoe e-commerce app with product management and backend API integration.
- Integrated the OpenAI API to generate product recommendations.
- Implemented secure JWT authentication and Razorpay payments.
- Optimized MongoDB aggregation pipelines to improve product query performance.

**Tech Stack:** React.js, Node.js, Express.js, MongoDB, OpenAI API, Razorpay, Tailwind CSS

## EDUCATION

**Thiruvalluvar College of Engineering and Technology** 2019 – 2021
*Master of Computer Applications (MCA)* *CGPA: 8.4/10*

**Shanmuga Industries Arts and Science College** 2016 – 2019
*Bachelor of Computer Applications (BCA)* *CGPA: 7.4/10*

## CERTIFICATIONS

**AI powered MERN Stack Developer** Apr 2026 – Jul 2026
*Coderead Academy*